

Assignment 1

Digital Signal Analysis
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Due date: 11:59pm on 18th March 2026

1. Solve all the question and submit a handwritten document.
2. Plagiarism will be penalised.
3. You can use any coding/programming language to complete ‘Coding’ questions.
4. Submit a pdf of the form `<roll no>_dsa1.pdf`
5. The submission should be a zip folder named with your rollnumber-dsa . Each coding question must be placed in a separate folder containing the corresponding code and results.

1 Questions

1. Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.
 - (a) $x(t) = 3 \cos(5t + \pi/6)$
 - (b) $x(n) = 3 \cos(5n + \pi/6)$
 - (c) $x(n) = 2 \exp[j(n/6 - \pi)]$
 - (d) $x(n) = \cos(n/8) \cos(\pi n/8)$
 - (e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3 \cos(\pi n/4 + \pi/3)$

2. (a) Show that the fundamental period N_p of the signals

$$s_k(n) = e^{j2\pi kn/N}, \quad k = 0, 1, 2, \dots$$

is given by $N_p = N/\text{GCD}(k, N)$, where GCD is the greatest common divisor of k and N .

- (b) What is the fundamental period of this set for $N = 7$?
- (c) What is it for $N = 16$?

3. **(Coding)** Consider the following analog sinusoidal signal:

$$x_a(t) = 3 \sin(100\pi t)$$

- (a) Sketch the signal $x_a(t)$ for $0 \leq t \leq 30$ ms.
 - (b) The signal $x_a(t)$ is sampled with a sampling rate $F_s = 300$ samples/s. Determine the frequency of the discrete-time signal $x(n) = x_a(nT)$, $T = 1/F_s$, and show that it is periodic.
 - (c) Compute the sample values in one period of $x(n)$. Sketch $x(n)$ on the same diagram with $x_a(t)$. What is the period of the discrete-time signal in milliseconds?
 - (d) Can you find a sampling rate F_s such that the signal $x(n)$ reaches its peak value of 3? What is the minimum F_s suitable for this task?
4. An analog signal $x_a(t) = \sin(480\pi t) + 3 \sin(720\pi t)$ is sampled 600 times per second.
- (a) Determine the Nyquist sampling rate for $x_a(t)$.
 - (b) Determine the folding frequency.
 - (c) What are the frequencies, in radians, in the resulting discrete-time signal $x(n)$?
 - (d) If $x(n)$ is passed through an ideal D/A converter, what is the reconstructed signal $y_a(t)$?
5. **(Coding)** *Sampling of sinusoidal signals: aliasing* Consider the following continuous-time sinusoidal signal

$$x_a(t) = \sin 2\pi F_0 t, \quad -\infty < t < \infty$$

Since $x_a(t)$ is described mathematically, its sampled version can be described by values every T seconds. The sampled signal is described by the formula

$$x(n) = x_a(nT) = \sin 2\pi \frac{F_0}{F_s} n, \quad -\infty < n < \infty$$

where $F_s = 1/T$ is the sampling frequency.

- (a) Plot the signal $x(n)$, $0 \leq n \leq 99$ for $F_s = 5$ kHz and $F_0 = 0.5, 2, 3$, and 4.5 kHz. Explain the similarities and differences among the various plots.
 - (b) Suppose that $F_0 = 2$ kHz and $F_s = 50$ kHz.
 - i. Plot the signal $x(n)$. What is the frequency f_0 of the signal $x(n)$?
6. **(Coding)** A discrete-time signal $x(n)$ is defined as

$$x(n) = \begin{cases} 1 + \frac{n}{3}, & -3 \leq n \leq -1 \\ 1, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) Determine its values and sketch the signal $x(n)$.
 - (b) Sketch the signals that result if we:
 - i. First fold $x(n)$ and then delay the resulting signal by four samples.
 - ii. First delay $x(n)$ by four samples and then fold the resulting signal.
 - (c) Sketch the signal $x(-n + 4)$.
 - (d) Compare the results in parts (b) and (c) and derive a rule for obtaining the signal $x(-n + k)$ from $x(n)$.
 - (e) Can you express the signal $x(n)$ in terms of signals $\delta(n)$ and $u(n)$?
7. Show that:
- (a) $\delta(n) = u(n) - u(n - 1)$
 - (b) $u(n) = \sum_{k=-\infty}^n \delta(k) = \sum_{k=0}^{\infty} \delta(n - k)$
8. (a) i. What is aliasing? What can be done to reduce aliasing?
- ii. Let $x(t) = \frac{1}{2\pi} \cos(4000\pi t) \cos(1000\pi t)$ be a continuous-time signal. Find the Nyquist rate and Nyquist interval for this signal.
- (b) A waveform, $x(t) = 10 \cos(1000t + \pi/3) + 20 \cos(2000t + \pi/6)$ is to be uniformly sampled for digital transmission.
- i. What is the maximum allowable time interval between sample values that will ensure perfect signal reproduction?
 - ii. If we want to reproduce 1 hour of this waveform, how many sample values need to be stored?